

Context- and Data-driven Satisfaction Analysis of User Interface Adaptations Based on Instant User Feedback

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Modern User Interfaces (UIs) are increasingly expected to be plastic, in the sense that they retain a constant level of usability, even when subjected to context (platform, user, and environment) changes at runtime. Adaptive UIs have been promoted as a solution for context variability due to their ability to automatically adapt to the context-of-use at runtime. However, evaluating end-user satisfaction of adaptive UIs is a challenging task, because the UI and the context-of-use are both constantly changing. Thus, an acceptance analysis of UI adaptation features should consider the context-of-use when adaptations are triggered. Classical usability evaluation methods like usability tests mostly focus on a posteriori analysis techniques and do not fully exploit the potential of collecting implicit and explicit user feedback at runtime. To address this challenge, we present an on-the-fly usability testing solution that combines continuous context monitoring together with collection of instant user feedback to assess end-user satisfaction of UI adaptation features. The solution was applied to a mobile Android mail application, which served as basis for a usability study with 23 participants. A data-driven end-user satisfaction analysis based on the collected context information and user feedback was conducted. The main results show that most of the triggered UI adaptation features were positively rated.

CCS Concepts: • **Human-centered computing** → **User interface management systems**; *Usability testing*;
 • **Software and its engineering** → *Software usability*.

Additional Key Words and Phrases: user interface adaptation, usability evaluation, context-aware computing

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1 INTRODUCTION

The user interface (UI) is a key component of any interactive software application and is crucial for the acceptance of the application as a whole. Modern applications have to consider contextual information to be functional and usable under various circumstances. The context-of-use defined by the triplet user, platform, and environment [8] characterizes and classifies these conditions. The ability of an application's UI to support heterogeneous contexts-of-use while maintaining usability is referred to as plasticity [7]. Due to different users, platforms, and environments, a wide variety of contexts-of-use must be considered in the design and development of plastic UIs.

Adaptive UIs have been promoted as a solution for plasticity due to their ability to automatically adapt to the context-of-use at runtime. Prior work in this field shows that adaptive user interfaces are a viable and useful option for coping with dynamically changing context-of-use situations [5].

While previous research in the field of adaptive UIs primarily focused on the engineering perspective, covering aspects like context or adaptation modelling, transformation and execution of adaptive UIs to reduce complexity in development, an in-depth usability evaluation of adaptive UIs regarding acceptance from the perspective of end-users is missing. Especially in the ubiquitous domain of mobile UI platforms, where dynamically changing context-of-use situations are usual, usability evaluation of UI adaptations is still a challenging task. For addressing this issue and analyzing the usability criteria end-user satisfaction of UI adaptations on mobile platforms, two research questions arise:

- (R1) Are UI adaptation features for mobile platforms accepted by the end users?
- (R2) Is it possible to identify situations in which UI adaptations are more or less likely to be accepted?

To find answers for these two questions, it is possible to make use of classical usability evaluation methods like usability tests, interviews or cognitive walkthroughs. However, these methods mostly focus on *a posteriori* analysis techniques and do not fully exploit the combined potential of implicit (collected context information) and explicit (collected user ratings) feedback at runtime. Therefore, a solution for usability testing of UI adaptations is needed, which tackles the following challenges:

- (C1) How can we test the acceptance of UI adaptation features despite the dynamic nature of adaptive UIs where context parameter and the UI itself are continuously changing at runtime?
- (C2) How can we test the acceptance of UI adaptation features at the very moment when the adaptations occur incorporating the current context-of-use?

To address challenges C1 and C2, we present an on-the-fly usability testing approach that integrates UI adaptation features and a user feedback mechanism into a mobile application. The developed solution enables us to continuously track various context information data and user feedback, e.g. whether the users like or dislike the triggered UI adaptations. To conduct a usability evaluation of adaptive UIs, we implemented an Android mail application enriched with diverse UI adaptation features and the described user feedback mechanism. The latter enables the users to grade the triggered UI adaptations at runtime while they are using the mobile application. Based on this implementation, we conducted a usability experiment with 23 participants who used and evaluated the mail application for seven days. The collected user feedback data enables us to investigate research questions R1 and R2.

The rest of the paper is structured as follows: In Section 2 we present the general solution idea for on-the-fly usability testing of UI adaptation features. Section 3 provides an overview of the concrete implementation of our adaptive mail application, which was used as a basis setup to perform our empirical experiment for evaluating adaptive UIs. In Section 4, we describe how the usability experiment was conducted. Section 5 presents main results and the interpretation of our

data-driven usability evaluation. In Section 6 we present related work. Section 7 concludes our work with an outlook on future work.

2 SOLUTION IDEA

A widespread technique to support UI plasticity is rule-based UI adaptation. Rule-based UI adaptation approaches such as [16] or [29] require a dedicated adaptation model consisting of UI adaptation rules that characterise how the UI is changed as a reaction to the changing context of use. In most cases, rule-based UI adaptation approaches foresee the usage of a rule engine like Nools¹ or Drools² to implement the adaptation logic by observing available facts of the context parameter and triggering the matching adaptation rules.

Our on-the-fly usability testing solution for UI adaptation features targets rule-based UI adaptation approaches and continuously monitors context information about context characteristics as well as collects instant user feedback about triggered UI adaptation features. To realize such an on-the-fly usability testing solution, we extended our existing rule-based UI adaptation approach [29] with capabilities to support continuous context monitoring and collect context-driven instant user feedback.

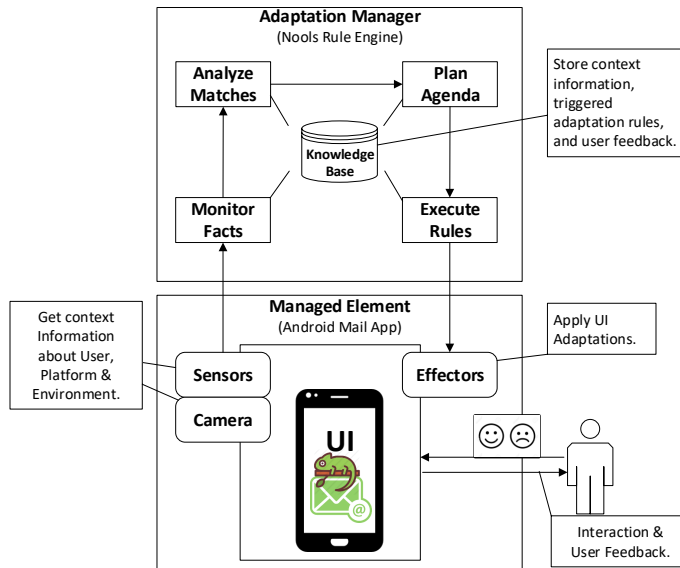


Fig. 1. Human-in-the-loop: On-the-fly usability testing of UI adaptation features

Figure 1 illustrates the main idea of our on-the-fly usability testing solution for UI adaptations. Based on IBM's MAPE-K loop [1], we have two main components *Adaptation Manager* and *Managed Element*. In our case, both main components are running on the same mobile platform. The *Adaptation Manager* is responsible for monitoring and adapting the *Managed Element*, which is the user interface of a mobile platform. For this purpose, the *Adaptation Manager* consists of five components. The *Monitor* component monitors various context information gathered through

¹<http://noolsjs.com/>

²<https://www.drools.org/>

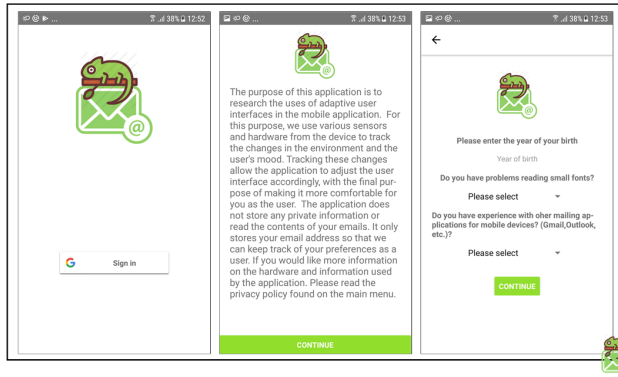


Fig. 2. Login (left), participation info (middle), and preliminary questionnaire (right) screens

integrated sensors and cameras on the mobile platform. The context information is then analyzed through the *Analyze* component and the *Plan* component decides whether UI adaptation rules match to the current context-of-use situation. If this is the case, the *Execute* component is responsible for executing the UI adaptation operations on the *Managed Element* through the *Effectors*. The *Effectors* represent concrete UI adaptation operations like layout or modality adaptation that are applied to the UI. Further technical details about the adaptation process are presented in our previous work where we describe each component related to context monitoring [27], analysis and plan [28], and execution of UI adaptations [29] in more detail.

While the described components are essential for supporting rule-based UI adaptation in general, further components and mechanisms are needed to support on-the-fly usability testing of UI adaptation features. To this end, we integrated an instant user feedback mechanism into the context monitoring and UI adaptation loop. The feedback mechanism allows users to explicitly rate the triggered UI adaptations. Users can give positive or negative feedback, but they can also ignore the feedback mechanism and concentrate on their main application task. As Figure 1 shows, there is a *Knowledge Base* which is responsible for storing all context information before and after a context change occurred (that lead to a UI adaptation triggering), all triggered adaptation rules, and the corresponding instant user feedback. Based on the stored information, it is possible to analyze the acceptance of UI adaptations based on the current context of the user and the user's feedback. As the *Knowledge Base* characterizes a key component and asset for the data-driven usability evaluation, all context information were collected in a central MySQL database.

3 IMPLEMENTATION

We implemented our on-the-fly usability testing solution for UI adaptation features based on an Android mail application using React Native³. We decided to use a mail application as email clients are one of the most frequently used applications on mobile platforms. People read and write mails while commuting to work, before going to sleep, while walking or watching TV or doing different other activities. The implemented email application relies on Gmail and provides full email services like send, receive, forward, delete and compose mails. In addition, a file manager is provided allowing users to download attachments and to open them using different applications on their mobile platform. Figure 2 shows the starting screens of the email application where login, participation info, and a preliminary questionnaire are depicted.

³<https://facebook.github.io/react-native/>

Beside the core mail application logic, inspired by [19], we implemented 28 UI adaptation features that cover different adaptation strategies like layout-, task-feature set- or modality-adaptation.

In the following, we first describe the main adaptation strategies and then explain each implemented adaptation feature in more detail. In general, the adaptation strategies can be divided into six main adaptation categories. *Task-feature-set* adaptations support UI adaptations by flexibly showing and hiding interaction elements of the UI. *Layout* adaptation enables UI adaptations that change the appearance of the UI by modifying presentation aspects like color, font (size), position, etc. *Modality* adaptation relates to changes where a switch from one interaction mode to another one is made (e.g., from the graphical to the vocal UI). It is also possible to have adaptation rules that deactivate previously executed UI adaptations. This type of rules belong to the category of *Deactivation* rules. Furthermore, we have identified a category for UI adaptations which are highly related to a specific domain and application area. We call this type of UI adaptations *Functional* adaptations as they provide an additional value for the underlying application which is going beyond the previously mentioned categories. Finally, it is important to note that adaptation features from the previous categories can be combined to a more complex adaptation feature, so called *Combined* adaptation.

In the following, we describe the implemented adaptation features and map them to the corresponding adaptation categories.

- Rule 1: Bad Mood. When the user is in a bad mood (detected through the face camera), a feedback text field is shown where the user can enter textual feedback. (*Task-feature*)
- Rule 2: Good Mood. When the user is in a good mood (detected through the face camera), a motivational quotation is shown. (*Task-feature*)
- Rule 3: Neutral Mood. When the user is in a neutral mood (detected through the face camera), rule 1 and 2 are deactivated and no pop-up menu appears. (*Deactivation*)
- Rule 4: Iconic UI. If the user has been using the app for a while (usage time), an iconic UI without text labels is shown (see screenshot on the left of Figure 3). (*Task-feature*)
- Rule 5: Minimized Navigation. If the user has been using the app for a while (usage time), extra information and text labels in the navigation bar are hidden. (*Task-feature*)
- Rule 6: Detailed UI. If the user is not experienced (based on preliminary questionnaire), additional information in the the UI and navigation are displayed. (*Task-feature*)
- Rule 7: Low Ambient Light. If the environmental lighting condition is low, adapt the UI color scheme and contrast to improve better readability in the dark. (*Layout*)
- Rule 8: Normal Ambient Light. If the environmental lighting condition is normal, adapt the UI color scheme and contrast to default mode. (*Layout*)
- Rule 9: High Ambient Light. If the environmental lighting condition is high, increase the UI contrast to improve readability. (*Layout*)
- Rule 10: Day Mode. When the day starts (clock timer), all UI elements are shown in the layout of the day mode. (*Layout*)
- Rule 11: Night Mode. When the day ends (clock timer), all UI elements are shown in the layout of the night mode. (*Layout*)
- Rule 12: Vocal UI. When the user is moving (accelerometer), all unread emails are read out using a text-to-speech (TTS) service (see screenshot on the right of Figure 3). (*Modality*)
- Rule 13: Impaired Vision. If the user has vision problems (preliminary questionnaire), the font size of the UI elements is increased. (*Layout*)
- Rule 14: Middle Aged User. Based on the entered age of the user (preliminary questionnaire), all UI elements are displayed in the default layout with a slightly bigger font size. (*Layout*)

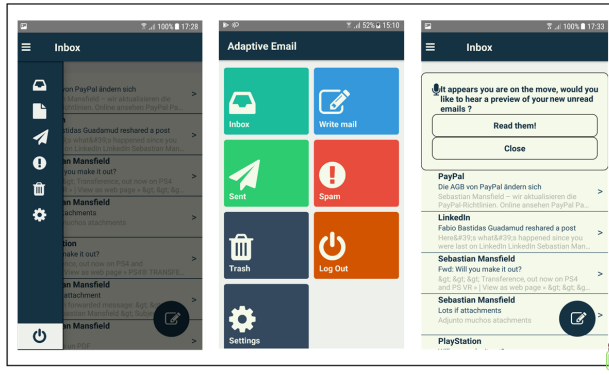


Fig. 3. UI Adaptations: "iconic" UI (left), "grid" UI (middle), vocal UI (right)

- Rule 15: Older User. Based on the entered age of the user (preliminary questionnaire), all UI elements are displayed in the grid layout with a bigger font size (see screenshot in the middle of Figure 3). (*Layout*)
- Rule 16: Younger User. Based on the entered age of the user (preliminary questionnaire), all UI elements are displayed in the default layout with the default font size. (*Layout*)
- Rule 17: Novice User. Based on the entered experience level of the user (preliminary questionnaire), the UI is adapted to the grid layout. (*Layout*)
- Rule 18: Expert User. Based on the entered experience level of the user (preliminary questionnaire), the UI is shown in the default layout. (*Layout*)
- Rule 19: Smartphone Mode. If the used device is a smartphone, a combined adaptation is executed where navigation and layout is optimized for smartphone devices. (*Combined*)
- Rule 20: Tablet Mode. If the used device is a tablet, a combined adaptation is executed where navigation and layout is optimized for tablet devices. (*Combined*)
- Rule 21: Driving Mode. When the user is driving (accelerometer), a combined adaptation is executed where the UI is displayed using the grid layout and additionally the text-to-speech service is activated for reading out mails. (*Combined*)
- Rule 22: Driving Mode Off. When the user is still (stopped or reached destination), the driving mode is deactivated. (*Deactivation*)
- Rule 23: Low Battery. When the battery level of the mobile platform is low, the UI is displayed in a black and white color scheme to save energy. (*Layout*)
- Rule 24: Normal Battery. When the battery level of the mobile platform is normal, rule 23 is deactivated. (*Deactivation*)
- Rule 25: Wi-Fi Connection. If the used device has a Wi-Fi connection, the emails are displayed in HTML format (if available) and attachments are automatically downloaded. (*Functional*)
- Rule 26: 3/4G Connection. If the used device has a 3/4G connection, the emails are displayed in HTML format (if available), but attachments are downloaded only on demand. (*Functional*)
- Rule 27: Slow Network. If the used device has a slow connection, the emails are displayed in text format and attachments are not automatically downloaded. (*Functional*)
- Rule 28: Vocal UI Off. When the user is still (accelerometer), rule 12 is deactivated. (*Deactivation*)

To realize the described UI adaptation features, we integrated the Nools rule engine into the mobile email application. The Nools rule engine is fed with context information from various integrated sensors and cameras of the mobile platform. For example, we use different sensors like

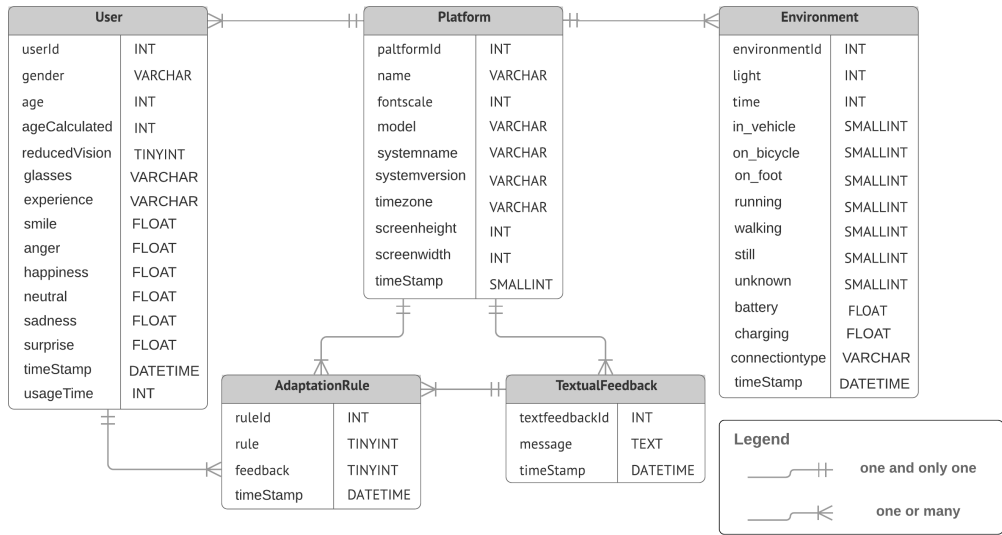


Fig. 4. Database schema: Stored context information as extended entity relationship (EER) diagram

ambient light, accelerometer, gyroscope, battery or network sensor. Furthermore, we use timer, device information or existing recognition services such as *Amazon Rekognition*⁴ to derive the user's emotion as well as estimated age or gender based on camera photos. A complete overview about the stored context information in the form of an extended entity-relationship model is depicted in Figure 4.

In order to avoid a cold start of the adaptation process, we have integrated a preliminary questionnaire form into our mobile email application that asks the users for their birth date, ability to read small fonts, and experience with other mailing applications for mobile devices. Based on this starting context information, the UI of the mobile email application can already adapt itself to a suitable starting configuration for the current user.

For realization of the feedback mechanism, we integrated a feedback prompt into the adaptive email application. The left screen in Figure 5 shows how the feedback prompt was placed in the mail application. On the top of the screen the feedback prompt is shown whenever context changes were detected that lead to UI adaptations. The triggered UI adaptations are explained in the feedback prompt (reason for adaptation and UI adaptation rule) and the user is able to provide feedback by clicking the positive or negative smiley indicating whether the user liked the UI adaptation or not. In some cases, for example, when the user is in a bad mood and the application detects this via camera, a feedback prompt in the form of a text field appears (see right screen in Figure 5) that allows the users to provide more detailed feedback.

4 USABILITY EXPERIMENT

During the usability experiment, the developed application was used for one week by 23 participants, who were recruited via an invitation mail. The test users were encouraged to use the application as often and in as many different contexts as possible. All users were made aware of the fact that their interaction with the application would be closely monitored (e.g. using facial recognition). In

⁴<https://aws.amazon.com/de/rekognition/>

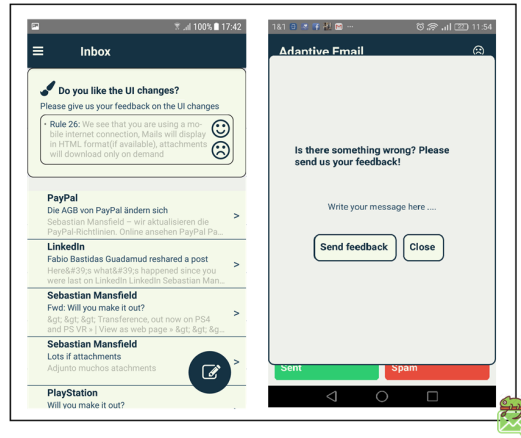


Fig. 5. Feedback prompt

the following, we present some facts and figures about the test user group and the encountered contexts.

The participant sample consisted of 10 male and 13 female users. This information was inferred from the facial recognition component. The facial recognition component also detected that 8 of the 23 users used reading glasses during the experiment.

Based on the prompt asking users for their date of birth when starting the application for the first time, the test users were between 22 and 55 years old. The majority of the users were in their twenties and thirties. From the 23 users, only 2 were 40 years or older. Eight of the users were between 30 and 40 years old. The rest of the users were younger than 30 years. Three users stated that they have problems reading small fonts while 20 answered that they do not have problems with small fonts. When asked for their experience with other mailing applications for mobile devices 13 users rated their experience level as high, 5 as medium, 3 as low and 2 of them stated that they have no experience.

5 USABILITY EVALUATION RESULTS

The data that was collected during the usability experiment allows us to evaluate the usability of the UI adaptation features in detail. We examine if the UI adaptations are accepted by end users (R1) and we investigate if the user context influences the acceptance of the different UI adaptations (R2). To investigate both research questions an essential requirement is that the application was indeed used in different contexts by the users. To make sure that this is the case, we first conduct a preliminary analysis of the collected data.

5.1 Preliminary Analysis

The goal of this preliminary analysis is to confirm that the participants did use the mobile application in different contexts. Figure 6 shows users and their aggregated usage time of the application. On average, each user spent almost 15 minutes in the application during the usability experiment. The user who used the application the most spent a total of 42 minutes in the application while the user who used the application the least only spent about two minutes in total in the application.

In Figure 7 it is shown how often the users gave feedback during each hour of the day. The users gave the most feedback in the time from 7 am to 8 pm. However, we also received 42 responses to UI

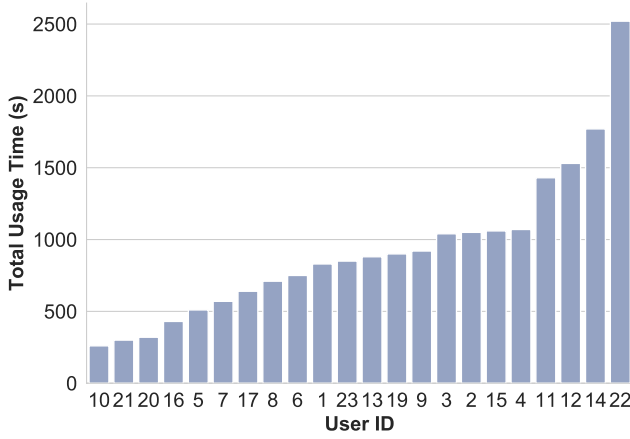


Fig. 6. Total usage time of the application for each user

adaptations during the rest of the day. This usage pattern provides several different context-of-use situations for our analysis.

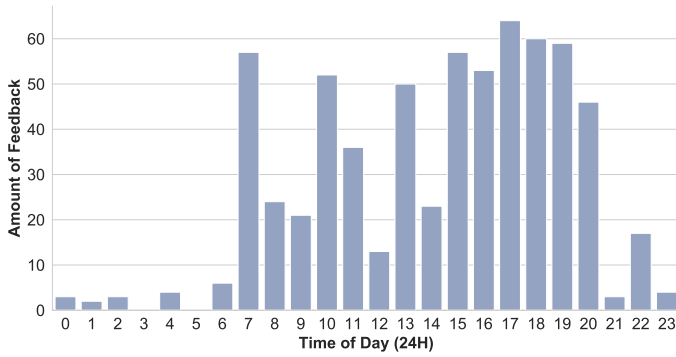


Fig. 7. Amount of given user feedback for each hour of the day

To assess the context-of-use, based on the used platform sensors, the application tries to predict the environmental circumstances (still, walking, in vehicle, etc.) of the user. In Figure 8 the amount of feedback that was given during different environmental conditions regarding movement is shown. We can infer that the application was used while commuting or in a fixed environment. All these context changes were used by the application to adapt the user interface. Furthermore, the figure 8 underlines the fact that user feedback about UI adaptations was received under dynamically changing context-of-use situations.

5.2 Research Question 1

In table 1, we can see some data about the context changes. There were 104404 detected context changes from all devices in the experiment. Of these, only 37465 triggered an adaptation by the rule engine. However, users gave feedback on the adaptation rules in only 663 cases. Every time an

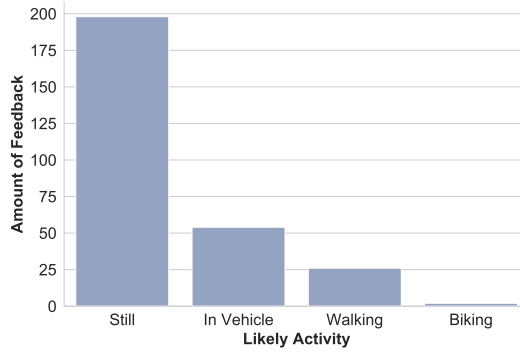


Fig. 8. Amount of user feedback collected for each predicted user activity. Only responses are shown for which the activity could be predicted with high certainty (280 cases).

Context changes		
Detected	Significant	Feedback
104404	37465	663

Table 1. Context change information

User feedback statistics			
Total	Positive	Negative	Percentage Positive
663	616	47	93

Table 2. User feedback

adaptation rule received feedback, the previous context additionally to the current context was saved.

For an initial evaluation of the research question R1, an analysis of the collected user feedback is done. In table 2 some descriptive statistics about the feedback provided by the users through the application are shown. In total, the users gave positive feedback in 616 cases and negative feedback in 47 cases. With about 93% of the feedback provided by users being positive this means that most of the user interface adaptations were liked by the users. However, this percentage should be interpreted carefully as the amount of collected feedback is low and the absence of explicit user feedback should not be interpreted as a positive result for end-user satisfaction in general.

Direct feedback

To complement the analysis, users where asked to give a short optional written feedback about the application and the adaptation rules. Although only seven users provided text feedback, their contributions can give us a more in-depth understanding of their feedback responses in the application and their perception of the usability of the application. Most of the users reported that they liked the application and its adaptations, and found them useful. Two users reported that although they liked the ability of the application to change colors in different context situations, the color schemes chosen by the adaptations were not to their liking. Two users reported that rule 22

Rule ID	Rule Name	Rule Description	Category	# Feedback Points	# Positive Feedback	# Negative Feedback	% Negative Feedback
1	Bad Mood	Feedback text field is shown.	T	10	9	1	10%
2	Good Mood	Quotation is shown.	T	13	10	3	23%
3	Neutral Mood	Deactivates rule 1 and 2.	D	21	17	4	19%
4	Iconic UI	Show less description info.	T	25	25	0	0%
5	Minimized Navigation	Navigation bar based on symbols.	T	0	0	0	-
6	Detailed UI	Display additional descr. info.	T	12	11	1	8%
7	Low Ambient Light	Adapt color scheme and contrast to low.	L	50	48	2	4%
8	Normal Ambient Light	Adapt color scheme to normal.	L	28	26	2	7%
9	High Ambient Light	Adapt color scheme and contrast to high.	L	2	2	0	0%
10	Day Mode	Show layout for day mode.	L	164	149	15	9%
11	Night Mode	Show layout for night mode.	L	33	30	3	9%
12	Vocal UI	Use text-to-speech (TTS) to read mails.	M	20	18	2	10%
13	Impaired Vision	Increase font size and layout.	L	3	3	0	0%
14	Middle Aged User	Adapt UI elements to middle size.	L	1	1	0	0%
15	Older User	Adapt to grid layout with big font size.	L	0	0	0	-
16	Younger User	Default layout and font sizes.	L	29	28	1	3%
17	Novice User	Adapt to grid layout.	L	6	6	0	0%
18	Expert User	Normal layout with additional fields.	L	0	0	0	0%
19	Smartphone Mode	Use smartphone optimized layout and nav.	C	19	19	0	0%
20	Tablet Mode	Use tablet optimized layout and nav.	C	2	2	0	0%
21	Driving Mode	Adapt to grid layout and use TTS.	C	16	16	0	0%
22	Drive Mode Off	Deactivate rule 21.	D	16	11	5	31%
23	Low Battery	Display black and white UI to save energy.	L	12	12	0	0%
24	Normal Battery	Deactivate rule 23.	D	5	5	0	0%
25	Wi-Fi Connection	Pre-download attachments automatically.	F	111	105	6	5%
26	3/4G Connection	Mails as HTML, no attachm. download.	F	62	60	2	3%
27	Slow Network	Mails as text, no attachm. download	F	3	3	0	0%
28	Vocal UI Off	Deactivate rule 12.	D	0	0	0	0%

Fig. 9. Overview: UI adaptation features and received user feedback

deactivated the driving mode at traffic lights or other temporary stops. This was found problematic by the users, hence the negative feedback.

5.3 Research Question 2

To gain more insights regarding the received feedback for the different rules, we created an overview of the UI adaptation features with the amount of received positive and negative feedback shown in Figure 9. We can see on the far left, the *Rule ID* of an adaptation feature, followed by the *Rule Name*, a short *Rule Description*, the *Category* (*Task-Feature* (T), *Layout* (L), *Modality* (M), *Functional* (F), *Combined* (C), *Deactivation* (D)) where the UI adaptation feature belongs to, the total number of feedback points, the number of positive feedback responses, negative feedback responses, and finally the percentage of negative feedback. Some of the UI adaptation features received no feedback at all, some received only positive feedback and some of them a combination of both positive and negative feedback. Four rules (14%) received no feedback from any user. This could be because the conditions for those rules were not met or users simply decided not to give feedback on these rules (users could ignore the feedback prompt and continue using the application). Eleven rules (almost 40%) received only positive feedback. No rules received only negative feedback.

From the 13 rules that received negative feedback, ten got mostly positive feedback with only 10% or less of the received feedback being negative. The rules that got most negative feedback are rule 22 (Drive Mode Off) with 31% negative feedback, rule 2 (Good Mood) with 23% negative feedback and rule 3 (Neutral Mood) with 19% negative feedback. Two of these rules, namely 3 and 22 are deactivation rules, meaning that the purpose of this adaptation rules is to deactivate the effects of rules whose condition is no longer active. Please note that looking only at the percentage of negative feedback can be in some cases a little misleading, as some rules received very little

feedback overall. In these cases, a single negative response can have a significant impact on the overall percentage value.

Relying on the shown results in Figure 9, by separately calculating the average amount of positive and negative feedback for each adaptation category, we derived a categorization of UI adaptation techniques according to their degree of end-user satisfaction. The results of this categorization are shown in Figure 10. As the amount of feedback data is low, especially regarding the number of negative feedback, and the received amount of feedback is not balanced for each category (see table column *Category* in Figure 9), the results should be cautiously interpreted. However, we can observe that *Combined* (C) UI adaptation features that were aggregated based on basic UI adaptation features such as rule 19 (neat combination of layout and navigation adaptation) have the largest impact on increasing user satisfaction. The second category which has also a big impact on increasing user satisfaction is *Functional* (F) UI adaptation. *Functional* UI adaptations, such as rule 25 are domain specific changes that provide additional value to the application for example by downloading the attachments if a high network connection is available. Good results regarding user satisfaction can be also reached by *Layout* (L) and *Task-Feature* (T) set adaptation rules. However, more attention has to be paid when it comes to the design of adaptive UIs incorporating *Modality* (M) changes and *Deactivation* rules. UI adaptations regarding *Modality* changes have potential for improving user satisfaction, but as the textual feedback by the users shows, many switches between two different modalities can also confuse the users and result in negative feedback. We observed that *Deactivation* rules for UI adaptations can have a negative influence on user satisfaction if timing and mapping to the current context-of-use is not well suited. The textual feedback by the participants shows, for example, that deactivation rule 22 was problematic when sudden context changes from the state driving to still were recognized and the vocal UI was deactivated. In this case, a warning message or better timing of UI adaptations was desired by the users.

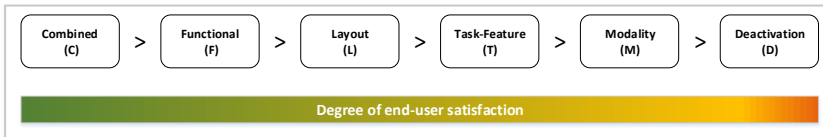


Fig. 10. Categorization of UI adaptation techniques according to their degree of end-user satisfaction

To gain a better understanding of the provided feedback, we also performed a graphical exploratory data analysis. We treated the feedback of the user to a UI adaptation as the dependent variable. The independent variables were composed of variables describing the context-of-use when the UI adaption was triggered and the context-of-use when the user feedback was given. The variables describing the context-of-use consist of the user, platform and environment variables shown in Figure 4. We graphically analyzed the interaction between the dependent and the independent variables with the goal to find patterns and anomalies. Since it is not possible to find any interesting interaction effects in case that (almost) all considered data points represent positive feedback, we only considered rules for which at least 4 negative responses exist. We note that due to the very limited amount of negative responses for all rules, it is not possible to draw any statistical significant conclusions based on the given data. However, we show based on the following example that our analysis allowed us to identify an unintended behaviour of the application based on the collected data.

Example: Analysis for Adaptation Rule 10

Rule number 10 is deactivating the night mode and activating the color scheme adaptation rule after the night color scheme is disabled. During the experiment, users gave feedback on this rule 164 times. From these responses, 149 were positive while only 15 were negative. During the exploratory data analysis we examined the relationship between the user feedback for this rule and the context variable describing the time of day using the visualization shown by Figure 11. It shows the distribution of positive and negative feedback responses via a density plot (created using kernel density estimation). We can see that while negative feedback was given at all times of the day, most of the negative feedback was received after 15:30. After a closer investigation, we derived the following explanation for this observation: Rule 10 correctly deactivates the night mode and displays that change to the user in the form of a feedback prompt. However, in cases where the application is used for the first time of the day in the evening, this might confuse users as they expect the activation and not the deactivating of the night mode. In fact, asking the user for feedback in these cases was not an intended behaviour of the application, because the user should have been only asked for feedback on recent context changes when starting the application. Our data-driven approach allowed us to identify this confusing and unintended behaviour of this adaptation rule.

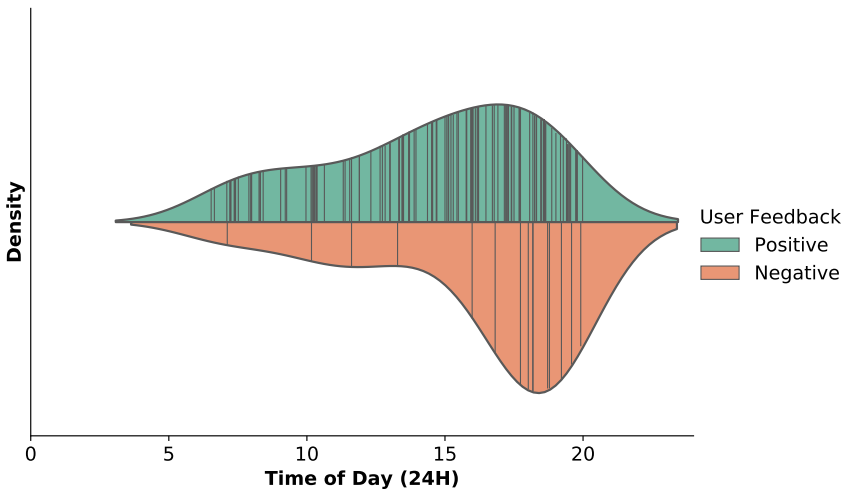


Fig. 11. Distribution of positive and negative feedback responses over the day. The green (top) and red (bottom) coloured areas show the density estimates of the time of day for positive and negative responses, respectively. The vertical lines represent the underlying datapoints.

5.4 Lessons Learned

Based on the conducted usability experiment and analysis of the results, we derived some lessons learned about specific UI adaptation categories as well as about the relationship between context-of-use and triggering UI adaptations.

One important finding is that users should only be asked for feedback on adaptations that concern context changes that the user is still aware of. UI adaptations regarding an outdated context change (which happened in the past) should just be discarded or applied without asking or informing the

user, who would not be able to understand it anyway as the context-of-use has probably already completely changed. This is especially the case when starting the application after a while. In this case, the user no longer knows the context of the last usage of the application.

Furthermore, a generic finding is that deactivation rules should have a (configurable) time window before they fire. This should be long enough to prevent unwanted reactions to temporary switches (a cloud, a traffic light, ...).

Regarding modality changes, it is important to prevent continuous changes in the interaction mode as users can be irritated. In this case, user feedback mechanisms about the allowed interaction mode or feedback about upcoming modality changes can be helpful to improve the user experience.

Likewise, as already presented in [4], task-feature set changes have a big potential to simplify the user interface. Regarding this adaptation category, it is essential to identify the right amount of information that is displayed on the screen, based on the current context-of-use.

Moreover, we observed that layout changes, in general, are useful to improve user satisfaction. However, as presented in [21] it is important to identify the individual and cultural preferences of the users as some adaptation rules were liked by a subset of the users and disliked by another subset of users.

Beside layout changes, the design of adaptive UIs can be improved when UI adaptations complement domain-specific functionalities of the targeted application.

Finally, we observed that a smart composition of previously mentioned basic UI adaptations to a combined UI adaptation has the largest potential to improve user satisfaction. An important and still open question regarding this aspect is how to combine different basic adaptation rules to a smart combined adaptation rule that satisfies the current context-of-use the most.

5.5 Limitations

In this section, we shortly discuss the limitations of this study and their impact on the validity of the results. Above all else, the small number of participants (and the low share of negative user responses) make it infeasible to draw any statistically significant conclusions from the experiment. In view of this fact, we are very careful to not make any definite claims regarding the interpretation of the results. Another limitation concerns the recruiting process of the participants of the experiment. The invitations were mostly sent to students, colleagues and friends. It must be assumed that the invited students and colleagues are (compared to the general population) rather tech-savvy. In addition, all groups are likely well disposed towards the organizers, potentially leading to a bias of the collected feedback data. With regard to a potential bias of the collected data, we have to underline the fact that users can ignore feedback questions and that the absence of explicit user feedback should not be interpreted as a positive result for end-user satisfaction. Therefore, the resulting percentage values should be rather seen as an indicator. Finally, an open point for discussion is the frequency of asking users for explicit feedback. Asking for feedback in this fine-grained manner can annoy the users and result in an intrusive evaluation method. However, it can be seen as a trade-off where one has to decide about the frequency of asking users and the amount of qualitative user feedback data that can be collected in addition to implicit interaction data. Although our informal interview with the participants showed that the separate design of the screen (feedback bar and "task specific" UI) mitigate the mentioned problem of annoyance to some extent, there is still room for improvement regarding this aspect.

6 RELATED WORK

Extensive research has been done in the field of adaptive user interfaces (UIs). There are several approaches, which adopt and apply the idea of adaptive UIs for different domains like health care systems [23], enterprise applications [3] or smart working environments [12].

As already described in prior work [10], usability evaluation of adaptive UIs is a challenging task that should be addressed by adequate methods and techniques. A literature review [25] is showing that mostly *questionnaires*, *interviews* and *data log analysis* methods are used for evaluating adaptive UIs. However, the authors argue that these methods are not fully suitable to investigate the usability of adaptive UIs due to quality of questionnaires, shallow state of think aloud protocols, or the need for triangulation of the data logs to derive useful information.

With regard to existing usability evaluation approaches, we can observe that various usability evaluation criteria such as ease of use, learnability, task performance time, accuracy, predictability, or satisfaction were investigated. In [20], for example, the authors investigate usability trade-offs for adaptive UIs with regard to ease of use and learnability. To this end, a user support concept was developed and applied to a context-aware mobile device with an adaptive UI. The approach was evaluated based on a usability test and main results show that the user support improved ease of use, but unexpectedly it reduced learnability of the adaptive UI. Moreover, in [14] for example, the authors examine the positive and possible adverse effects of adaptive UIs in the context of an in-vehicle telematic system. For this purpose, they conduct a usability experiment with different participants to measure the task performance time of the users using the proposed adaptive UI. Other studies such as [11] are examining the relative effects of predictability and accuracy on the usability of adaptive UIs. The results of this study show, for example, that increasing predictability and accuracy lead to strongly improved satisfaction.

Beside above mentioned classical usability evaluation methods, prior work in the field of plastic user interfaces also investigated the question how the notion of mappings as promoted by *Model Driven Engineering (MDE)* can be exploited to control UI adaptation according to explicit usability criteria. In [24], for example, the authors present a formal mapping between source and target models specifying the usability properties that are preserved when transforming source models into target models. However, the applicability of such formal methods in real application scenarios is quite difficult as ergonomic criteria for ensuring usability may be inconsistent, and as a result, require difficult trade-offs [24].

More recent approaches from the area of mixed-initiative interfaces [6], such as *CrowdAdapt* [18], make use of a combination of end-user development and crowd-sourcing for context-aware adaptation of UIs. Furthermore, [15] suggests to incorporate user feedback via a promoting/demoting technique to further improve UI adaptations. While this approach mainly focuses on UI adaptation improvement based on a combination of user feedback and machine learning, the idea of collecting instant user-feedback can be also used to test the end-user satisfaction of UI adaptations. Inspired by this work, we developed our presented on-the-fly usability testing solution that is especially focusing on rule-based UI adaptation approaches.

As research questions R1 and R2 are indicating, the leading research topic for our work is to investigate the acceptance of adaptive UIs in mobile applications by creating a real-life application and collecting feedback from the users. In this sense, our work differs from current research done in the field. For a systematic comparison of our solution with the existing related work, we used the following four main criteria:

- *End-user Satisfaction Analysis*: Whether the approach considers usability evaluation regarding end-user satisfaction.
- *Instant User Feedback*: Whether the approach incorporates instant user feedback for usability evaluation.
- *Usability Test at Runtime*: Whether the approach addresses usability evaluation at runtime while the user interacts with the system.

The tabular overview in Figure 12 shows the main results of the comparison.

	Criteria		
	End-user Satisfaction Analysis	Instant User Feedback	Usability Test at Runtime
Legend  Completely fulfills  Partially fulfills  Does not fulfill			
AppEcho (Seyff et al.)			
AUE (Feuerstack et al.)			
AUI Mobile (Mitrovic et al.)			
Media Maps (Wesson et al.)			
MOCCA (Reinecke et al.)			
Mining Minds (Hussain et al.)			
OSApp (Abusair et al.)			
RBUIS (Akiki et al.)			
On-the-fly Usability Testing			

Fig. 12. Comparison of our on-the-fly usability testing solution with the existing work

AppEcho [22] is a user-driven, in situ feedback approach for mobile platforms and applications. The authors present a mobile feedback approach, which enables users to document individual feedback on mobile systems in situ. The collected information can then be evaluated and used as new requirements by developers. While this approach explicitly targets mobile platforms and uses instant user feedback mechanisms as in our solution, it is only partially focusing on end-user satisfaction analysis. With this regard, the *AppEcho* approach is providing a more generic feedback mechanism which is not explicitly focusing on the evaluation of UI adaptations and their effect on user satisfaction based on the context-of-use. As the *AppEcho* approach aims to provide a generic feedback mechanism to collect a large amount of individual user feedback about mobile applications, it is not designed with the purpose to support usability tests at runtime.

Another related approach, called *Automated Usability Evaluation* (AUE) of model-based interactive systems, is presented in [9]. In this paper, the authors describe an approach to efficiently evaluate the usability of an interactive application that has been realized to support various platforms and modalities. By using a model-based runtime environment incorporating a mental model of the end-user, the authors aim to reduce the evaluation effort by automating parts of the testing process for various combinations of platforms and user groups. While this approach supports usability testing at runtime and also targets mobile platforms, it is only partially covering the aspect of instant user feedback. As the user is not looped in to the real interaction scenario and is represented through a model, an explicit feedback of the real end-users cannot be collected. In contrast to our solution, this approach is also not focusing on the usability aspect end-user satisfaction.

AUI Mobile [17] is a further approach which is investigating the application of adaptive user interfaces for mobile platforms. In this work, the authors propose a solution for the adaptation of graphical user interfaces by using a mobile agent system. Applicability of this approach is shown based on a mobile currency converter and a survey application. This approach mainly focuses

on the application of adaptive UIs for mobile platforms. However, this work is not focusing on usability evaluation, instant user feedback or usability testing of UI adaptation features at runtime.

A usability evaluation study of adaptive UIs for mobile platforms is presented in [26]. In this work, the main research question is whether adaptive UIs improve the usability of mobile applications. For this purpose, the authors discuss several simple types of UI adaptations based on a case study dealing with *Media Maps*. Based on this case study, they conduct a usability test with 20 participants with a post-test satisfaction questionnaire. The main results of this usability study show that adaptive UIs have potential to improve mobile applications regarding accuracy and end-user satisfaction. Compared to our solution, the described usability study does not incorporate instant user feedback nor establishes a way to efficiently test the end-user satisfaction rate of individual users based on their current context-of-use.

A deeper analysis of various usability aspects such as performance, perceived usability, and aesthetics is presented in the *MOCCA* [21] approach, which is focusing on culturally adaptive user interfaces. In this work, the authors argue that it is not feasible to design one interface that appeals to all users of an increasingly global audience. Instead, they propose to design culturally adaptive systems, which automatically generate personalized interfaces that correspond to cultural preferences. Based on a usability test with 41 participants of different cultural backgrounds, they demonstrate the benefit of the approach by showing that the majority preferred their culturally personalized interfaces to a non-adapted version. Moreover, this study shows that the participants were able to work 22% faster with the culturally adapted interface, showing that the approach improves both the performance as well as the user satisfaction. While the *MOCCA* approach analyzes the usability of culturally adaptive UIs for different platforms such as mobile or desktop and also focuses on the aspect of user satisfaction, it is not incorporating instant user feedback for efficiently testing the usability of different UI adaptations at runtime. However, this is essential for efficiently conducting long-term usability studies about adaptive UIs with a large amount of participants.

The *Mining Minds* [13] platform is a further approach which supports UI adaptation and provides a detailed usability evaluation based on context and user experience. The authors of this work present a domain and platform independent model-based methodology for devising adaptive user interfaces. The methodology is implemented as a tool capable of adapting the user interface based on contextual factors. As in our solution, the *Mining Minds* platform supports various UI adaptation techniques like task-feature set, layout, and modality adaptation for different platforms including mobile devices. In addition, an instant user feedback mechanism is integrated in this approach. However, the collected user feedback is more general and does not allow to rate specific UI adaptations in a specific context-of-use. Furthermore, the authors do not provide evaluation results based on the collected user feedback data, but rather present the results of a usability experiment with 32 participants based on a questionnaire. Although the *Mining Minds* platform is the most similar approach to ours, the main difference is that our approach supports a fine grained rating for triggered UI adaptation features reflecting the current context-of-use at runtime. Also in our solution, we directly use the collected context and corresponding user feedback data to perform a data-driven satisfaction analysis of UI adaptation features.

In [2], the authors present a context-aware adaptation approach for mobile applications driven by software quality and user satisfaction. The proposed approach uses a reward formula, containing the notion of software availability and the user's feedback to determine the best adaptation that needs to be applied in a given context. The feasibility of this approach is shown based on a mobile application, called *Off Site Art OSApp*. As a result of their approach the authors provide a reward formula by describing the user satisfaction model and user-perceived service availability model. While this approach covers end-user satisfaction analysis and targets mobile platforms, it does not

focus on instant user feedback and usability testing at runtime. However, the introduced reward formula in this work could be used to also assess the usability of adaptive user interfaces for mobile platforms.

Role-Based UI Simplification (RBUI) [4] is a further approach for supporting adaptive UIs. The authors of this work define UI simplification as a mechanism for increasing usability through adaptive behaviour by providing users with a minimal feature-set and an optimal layout based on the context-of-use. The *RBUI* approach mainly focuses on simplification of enterprise applications accessed via a desktop computer, but also considers mobile device usage. The authors evaluated the approach using an online interactive survey with a UI pair composed of an initial and a simplified UI. The main results of the user study with 25 participants show that simplifying enterprise application UIs based on roles improves user satisfaction and efficiency. Although user feedback is partially integrated in the *RBUI* approach by allowing users to *Apply and Keep* and *Apply Once* of UI simplification operations, it is not used for explicitly testing the acceptance of specific UI adaptation features at runtime. In contrast to our solution, *RBUI* is also not directly mapping the current context-of-use to the provide feedback by the users.

In summary, the existing approaches and usability evaluation methods are not fully addressing the analysis of end-user satisfaction of adaptive UIs for mobile platforms by combining instant user feedback and usability testing of UI adaptation features at runtime. Therefore, our presented solution contributes a novel method for testing and analysing adaptive UIs.

7 CONCLUSION AND OUTLOOK

This paper presents a novel solution for testing the usability of UI adaptations for mobile platforms with a special focus on end-user satisfaction. To overcome the problems of existing approaches for evaluating user satisfaction of adaptive UIs, we introduce an on-the-fly usability testing solution that incorporates instant user feedback and the corresponding context-of-use. In this way, our solution enables us to continuously track various context information about the user, platform, and environmental characteristics and user feedback, e.g. if the users accept the triggered UI adaptations or not. The developed solution was applied to an Android mail application and served as a basis to conduct an experiment with 23 participants. Main results of our usability study show that most of the provided UI adaptation features were liked by the users. In addition, we were able to find patterns which UI adaptation features were more likely to be accepted by the users. On the one hand, we identified that combined UI adaptation features which were aggregated based on basic UI adaptation features have the biggest impact on increasing user satisfaction. On the other hand, we observed that deactivation rules for UI adaptations can have a negative influence on user satisfaction if the timing and the mapping to the current context-of-use is not well implemented. In summary, we have presented a novel usability testing solution which combines implicit (collected context information) and explicit (user ratings) feedback to evaluate end-user satisfaction of adaptive UIs. The presented solution can be transferred and used to evaluating adaptive UIs that employ a rule-based approach to UI adaptation.

Future experiments, especially long-term evaluation studies should be done with a larger group of participants to collect even more data about the context-of-use and instant user feedback to support evidence for our first findings. We are confident that given the data of a larger user test group, it is possible to identify statistical significant interactions between the context-of-use and the user feedback when using more sophisticated techniques such as regression analysis. Future experiments should also cover a direct comparison between the adaptive and non-adaptive version of the UI. For this purpose, we are planning to support undo operations for the triggered UI adaptations to allow for an easier comparison of the UI before and after the adaptation. In addition to that, we plan to establish feedback mechanisms where users can differentiate between negative feedback

for undesirable, and desirable but wrongly (according to the specific user) triggered adaptations. In ongoing work, we are also extending the presented solution by mechanisms to support the evaluation of further usability criteria like effectiveness or efficiency. Further improvements could also address the reusability of implemented features like UI adaptations or feedback mechanism to ensure an easy adoption for further applications in other domains. For future work, we have several ideas on how to extend the approach to improve the solution for evaluating adaptive UIs and, as a consequence, how to improve the quality of UI adaptation features. Firstly, a mutation analysis approach can be integrated in our solution to intersperse "bad UI adaptations", so called mutants in the application. This way, a more objective rating from end-user perspective can be reached, as they are not always shown useful adaptation features. In addition, it is possible to use implicit feedback mechanisms (e.g. shutdown of app, mood, usage time etc.) which are already tracked in our approach to further improve UI adaptations. In some cases, dependent on the application domain, such as in the presented mail application, it is also possible to adapt the UI based on the content of the UI elements, like the mail content itself. Furthermore, the collected data about user feedback and context-of-use can be used and combined with machine learning techniques to further improve UI adaptations.

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